

Experiment 4: Basic Image Handling and Processing Operations

Aim

To read an image using Python and dilate the image using the Dilate function.

Algorithm

1. Import the OpenCV (cv2) and NumPy libraries.
2. Read the input image using cv2.imread().
3. Create a kernel using NumPy.
4. Apply dilation using cv2.dilate().
5. Display the original and dilated images.
6. Wait for a key press and close all display windows.

Python Code

```
1  import sys
2  from pathlib import Path
3  import cv2
4  import numpy as np
5
6
7  def dilate_image(input_path: str, show_windows: bool = False) -> None:
8      image = cv2.imread(input_path)
9
10     if image is None:
11         raise FileNotFoundError(f"Could not read image: {input_path}")
12
13     kernel = np.ones((5, 5), np.uint8)
14     dilated_image = cv2.dilate(image, kernel, iterations=1)
15
16     input_file = Path(input_path)
17     output_path = input_file.with_name(f"{input_file.stem}_dilated{input_file.suffix}")
18     cv2.imwrite(str(output_path), dilated_image)
19
20     print(f"Input image read from: {input_path}")
21     print(f"Dilated image saved as: {output_path}")
22
23     if show_windows:
24         cv2.imshow("Original Image", image)
25         cv2.imshow("Dilated Image", dilated_image)
26         cv2.waitKey(0)
27         cv2.destroyAllWindows()
28
29
30  def main() -> None:
31      input_file = sys.argv[1] if len(sys.argv) > 1 else "input.jpg"
32      show_windows = "--show" in sys.argv[1:]
33      dilate_image(str(Path(input_file)), show_windows=show_windows)
34
35
36  if __name__ == "__main__":
37      main()
```

INPUT



OUTPUT



Result

The input image was successfully read using Python, and the image was dilated using the Dilate function successfully.